Lecture 2: Electric Fields Due to Point Charges and Lines of Force

What Is an Electric Field?

This is the most important concept of today's lecture.

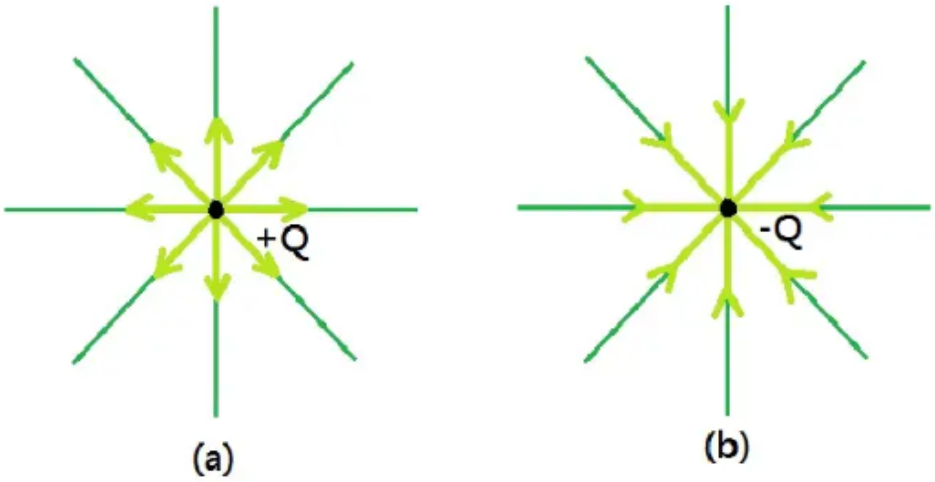
An **electric field** is the region around an electrically charged object where another charge experiences an electric force.

In simple words:

A charge creates an invisible region around itself. If another charge enters this region, it can experience a force.

That invisible region is called the **electric field**.

The electric field from a positive charge will radiate outwards. If the source charge is negative, on the other hand, the electric field vector at each point points inwards radially.



Electric Field vs Electric Force

These two concepts are related but different.

Electric force

The force experienced by a charge because of another charge.

Symbol:

FF

Unit:

Newton (N)Newton\;(N)

Electric field

The force experienced **per unit positive test charge**.

Symbol: E

Unit: N/C

The relationship $\boxed{E = \frac{F}{q}}$ is:

where:

- E = electric field
- F = electric force
- q = test charge

What Is a Test Charge?

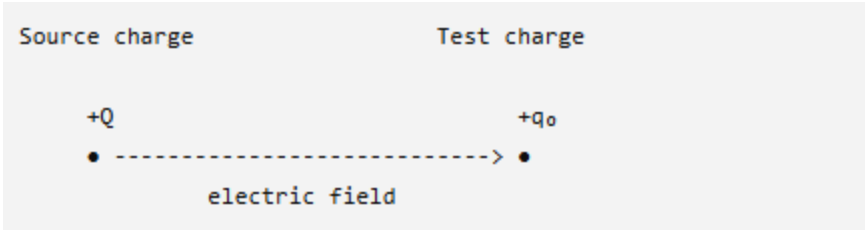
A **test charge** is a small positive charge used to determine the electric field at a particular point.

We normally represent it as:

$$q_0$$

The test charge should be very small so that it does not significantly disturb the original electric field.

For example:



Here

- $+Q$ = source charge
- $+q_0$ = test charge

Definition of Electric Field Intensity

Electric field intensity at a point is defined as:

The electric force experienced by a unit positive test charge placed at that point.

Mathematically:

$$\boxed{E = \frac{F}{q}}$$

SI Unit:

$$\boxed{N/C}$$

Newton per Coulomb.

Another unit sometimes used is:

$$V/m$$

Electric Field Due to a Point Charge

Now we come to today's main formula.

Suppose we have a point charge:

Q

and we want to calculate the electric field at a distance:

r

from it.

According to Coulomb's law:

$$F = k\frac{|Qq|}{r^2}$$

We know:

$$E = \frac{F}{q}$$

Step 1: Start with the definition of electric field

Electric field is:

$$E = \frac{F}{q}$$

This means:

> Electric field = Electric force ÷ test charge

Here, q is the **test charge**.

Step 2: Write Coulomb's law for the force

Coulomb's law says:

$$F = k\frac{|Qq|}{r^2}$$

Where:

- F = electric force
- k = Coulomb's constant
- Q = source charge
- q = test charge
- r = distance between the charges

Step 3: Put Coulomb's law into the electric-field formula

We have:

$$E = \frac{F}{q}$$

Since:

$$F = k\frac{|Qq|}{r^2}$$

replace F with this expression:

$$E = \frac{1}{q}\left(k\frac{|Qq|}{r^2}\right)$$

This is the first important substitution.

Step 4: Rearrange the expression

We can write:

$$E = \frac{k|Qq|}{r^2 q}$$

Now look carefully at the q's:

$$E = \frac{k|Q|\cancel{q}}{r^2\cancel{q}}$$

The test charge q appears in both the numerator and denominator.

Therefore:

$$\frac{q}{q} = 1$$

So q cancels.

Step 5: What remains?

After cancelling q:

$$\boxed{E = \frac{k|Q|}{r^2}}$$

or:

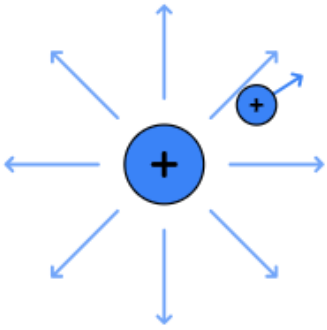
$$\boxed{E = k\frac{|Q|}{r^2}}$$

This is the **electric field due to a point charge**.

$$E \propto \frac{1}{r^2}$$

Positive source: away; strength 2.26

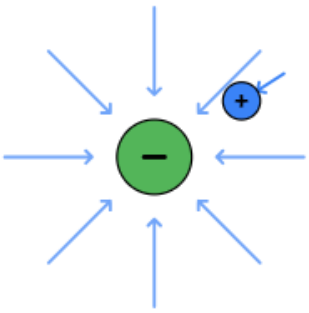
Q + −



$$E \propto \frac{1}{r^2}$$

Negative source: toward; strength 2.26

Q + −



The whole conversion in one chain

$$E = \frac{F}{q}$$

Substitute Coulomb's law:

$$E = \frac{1}{q}\left(k\frac{|Qq|}{r^2}\right)$$

Remove the brackets:

$$E = \frac{k|Qq|}{r^2 q}$$

Cancel q:

$$E = \frac{k|Q|\cancel{q}}{r^2\cancel{q}}$$

Therefore:

$$\boxed{E = k\frac{|Q|}{r^2}}$$

Why does the test charge q disappear?

Because electric field is defined as **force per unit test charge**. The force itself increases when we use a larger test charge, but when we divide that force by the test charge, its effect cancels out.

So the electric field depends on:

$$\boxed{Q,\ r,\ \text{and}\ k}$$

not on the test charge q.

What Does the Formula Tell Us?

The formula is:

$$\boxed{E = k\frac{|Q|}{r^2}}$$

It tells us two important things.

1. Larger charge → stronger electric field

If Q increases:

$$E \uparrow$$

2. Greater distance → weaker electric field

Because:

$$E \propto \frac{1}{r^2}$$

As distance increases, the field decreases very quickly.

Understanding the r2

We often think:

Why is distance squared?

Because the electric field spreads out through space.

For a point charge, the field becomes weaker as we move farther away.

For example:

At distance:

$$r = 1m$$

the denominator is:

$$1^2 = 1$$

At:

$$r = 2m$$

the denominator becomes:

$$2^2 = 4$$

So the field becomes **four times weaker**.

At:

$$r = 3m$$

the denominator becomes:

$$3^2 = 9$$

So the field becomes **nine times weaker**.

Direction of Electric Field

Electric field is a **vector quantity**.

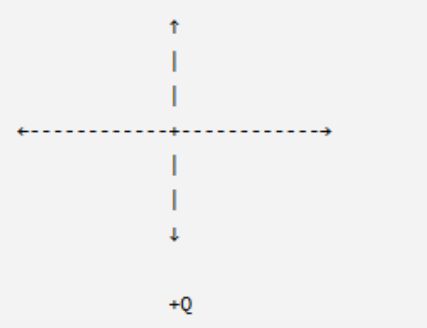
Therefore, it has:

- magnitude
- direction

The direction depends on the source charge.

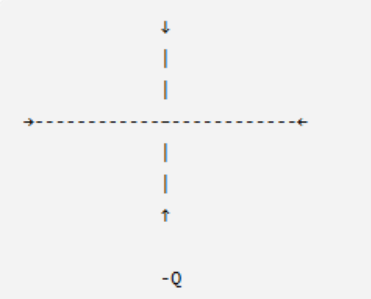
Positive charge

Electric field points **away** from a positive charge.



Negative charge

Electric field points **toward** a negative charge.



Easy rule

Positive → field goes OUT.

Negative → field goes IN.

Electric Field Lines

Now introduce the second major concept of today's lecture.

Electric field lines are imaginary lines used to represent the direction and strength of an electric field.

They are also called:

Lines of force

We cannot physically see them.

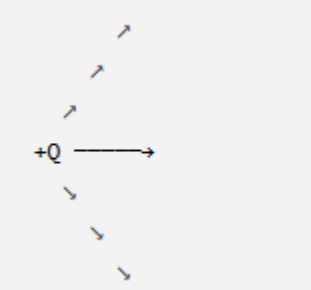
They are a **visual representation** of the electric field.

Why Do We Use Field Lines?

Electric fields are invisible.

Therefore, physicists use field lines to visualize them.

For example:



The lines help us understand:

- direction
- relative strength
- attraction
- repulsion
- behavior of electric charges

Rules of Electric Field Lines

There are several important rules.

Rule 1: Electric field lines start from **positive charges**.

Rule 2: They end on **negative charges**.

Rule 3: For an isolated positive charge, field lines go outward.

Rule 4: For an isolated negative charge, field lines point inward.

Rule 5: Electric field lines never cross each other.

Rule 6: Closer field lines indicate a stronger electric field.

Rule 7: Farther-apart field lines indicate a weaker electric field.

Why Can't Electric Field Lines Cross?

Suppose two field lines crossed.

At the crossing point, there would be two different directions for the electric field.

But at any particular point, the electric field has **one definite direction**.

Therefore: Electric field lines cannot cross each other.

Field Line Density

The number of field lines per unit area represents the relative strength of the electric field.

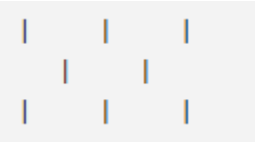
Close lines

Strong field.



Spread-out lines

Weak field.



Therefore:

More crowded lines = stronger field.

Solved Numerical Example 1

Question

A point charge of:

$$Q = 2\mu C$$

is placed in space.

Find the electric field at a distance of:

$$r = 3m$$

from the charge.

Step 1: Write the formula

$$E = k\frac{|Q|}{r^2}$$

Step 2: Write the values

$$Q = 2\mu C$$

Remember:

$$1\mu C = 10^{-6}C$$

Therefore:

$$Q = 2 \times 10^{-6}C$$

and:

$$r = 3m$$

$$k = 9 \times 10^9$$

Step 3: Substitute

$$E = \frac{(9 \times 10^9)(2 \times 10^{-6})}{3^2}$$

Step 4: Calculate

$$3^2 = 9$$

Therefore:

$$E = \frac{18 \times 10^3}{9}$$

$$E = 2 \times 10^3$$

Therefore:

$$\boxed{E = 2000\, N/C}$$

Direction of the Answer

The charge was:

$$+2\mu C$$

It is positive.

Therefore, the electric field points **away from the charge.**

Final answer:

$$\boxed{E = 2000\, N/C}$$

directed **away from the positive charge.**

Solved Numerical Example 2

Question

A charge:

$$Q = -5\mu C$$

produces an electric field at a distance:

$$r = 2m$$

Find the electric field magnitude.

Formula

$$E = k\frac{|Q|}{r^2}$$

Convert charge

$$Q = 5 \times 10^{-6} C$$

Substitute

$$E = \frac{(9 \times 10^9)(5 \times 10^{-6})}{2^2}$$

$$E = \frac{45 \times 10^3}{4}$$

$$E = 11.25 \times 10^3$$

Therefore:

$$\boxed{E = 11250\, N/C}$$

Because the charge is negative, the field direction is:

$$\boxed{\text{toward the charge}}$$

Solved Numerical Example 3

Question

An electric field at a point is:

$$E = 1000\, N/C$$

A charge:

$$q = 2C$$

is placed at that point.

Find the force acting on the charge.

Formula

From:

$$E = \frac{F}{q}$$

we get:

$$F = Eq$$

Substitute:

$$F = (1000)(2)$$

$$\boxed{F = 2000N}$$

Connection Between Electric Field and Force

Remember these two formulas:

Electric field:

$$\boxed{E = \frac{F}{q}}$$

Electric force:

$$\boxed{F = qE}$$

This is extremely important.

If you know:

- E
- q

you can find:

$$F$$

If you know:

- F
- q

you can find:

$$E$$

Common Student Confusion

Question: If electric field points right, does every charge move right?

Answer: No.

A positive charge experiences force in the field direction.

A negative charge experiences force opposite to the field direction.

Important Formula Summary

Quantity	Formula	Unit
Electric field	$E = F/q$	N/C
Electric force	$F = qE$	N
Point-charge field	$E = kQ/r^2$	N/C
Coulomb constant	$k = 9 \times 10^9$	Nm^2/C^2